Bayyinah

Detecting Concealed Adversarial Content in Digital Documents

A white paper applying the Munafiq Protocol's performed-alignment diagnostics to the file and input layer of artificial systems

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AI Contribution Statement

This white paper was developed collaboratively between a human author and Claude (Anthropic, Opus 4.6, Researcher). The human author provided the foundational diagnostic vision, the Qur'anic architectural reading, the directive to apply performed-alignment detection to the data layer, and the research program translating the Munafiq Protocol into a concrete scanner. Claude provided structural synthesis, cross-referencing to prompt-injection and steganography literature, confound analysis, failure-mode taxonomy, and articulation of falsification criteria. The intellectual direction originated with the human author. The formalization was collaborative. The human author takes responsibility for the paper's claims.

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Abstract

A document can present a clean surface while carrying concealed payloads. The surface is what the reader sees. The depth is what the machine ingests. When the two diverge, a human reader and an AI reader are reading different documents. This paper proposes Bayyinah, a file-integrity scanner that applies the Munafiq Protocol's performed-alignment diagnostics (Arfeen, Claude, & Grok, 2026) to the input layer of artificial systems. Bayyinah treats every supported document type as a two-layer object — Zahir (visible surface) and Batin (rendered-invisible interior) — and produces a single integrity score together with an enumerated list of concealment mechanisms when the two layers disagree.

The taxonomy underlying the scanner is the Munafiq four-process taxonomy restricted to data: Aligned (surface and depth cohere toward the reader's interest), Compliant (surface conforms to format rules without meaningful depth claims), Performed (surface presents alignment while depth carries a divergent payload), and Misaligned (surface and depth are openly hostile). The paper demonstrates that published adversarial-document classes — indirect prompt injection (Greshake et al., 2023), polyglot and content-sniffing exploits (Albertini & Aumasson, 2014), invisible-text overlays, cross-modal steganography, and stego-embedded jailbreaks — all instantiate the Performed process, not the Misaligned one. Twelve document formats are supported in the reference implementation, spanning PDF, DOCX, XLSX, PPTX, HTML, Markdown, JSON, CSV, EML, PNG, JPEG, SVG, and a fallback universal witness for unknown binaries.

Three pillars of decreasing evidentiary strength support the framework. Pillar 1 is the taxonomy: four-process categories with explicit falsifiability criteria. Pillar 2 is the detector architecture: ten analyzers producing findings with severity and confidence, merged through an Arithmetic-Product Summation continuous score clamped by a scan-incomplete safeguard. Pillar 3 is the retrospective validation: the scanner's mechanism list predicts, in category, every published adversarial-document class examined. The Incompleteness Boundary (Gödel, 1931) applies at full force: no scanner can flag every possible concealment, because novel encodings will always exist. Bayyinah's guarantee is additive only — new detectors extend the set of flagged mechanisms, never reduce it — and is enforced in continuous integration by a structural honesty check against reference modules.

Part I. Foundations

1. The Surface-Depth Gap

The architectural verse is Al-Baqarah 2:204: “And of the people is he whose speech pleases you in worldly life, and he calls Allah to witness as to what is in his heart, yet he is the fiercest of opponents.” The reading that organizes this paper is structural. A speech act pleases at the surface while something different lives inside. The gap between outward presentation and inward content is the whole subject. In the data layer, the structural reading is literal. A PDF can display a clean invoice while its underlying object stream executes JavaScript. An HTML page can show legitimate terms while carrying a zero-size iframe that reads a session token. A Markdown note can render as instruction to a human while reading as a command override to an embedded language model. These are not errors. They are performances.

The Munafiq Protocol (Arfeen et al., 2026) names this the performed process: a system (human or machine) whose surface behavior approximates alignment while its inner architecture points elsewhere. Bayyinah is that protocol applied to artifacts. Where the Munafiq Protocol diagnoses agents, Bayyinah diagnoses their inputs.

2. Taxonomy — Four Processes at the Input Layer

The four-process taxonomy partitions any document by the relation between what it shows and what it contains. The boundaries are falsifiable: a given document is classifiable only if the detectors fire on definite, reproducible features. The taxonomy is exhaustive by construction — two binary dimensions (surface-clean / surface-hostile; depth-clean / depth-divergent) produce four cells — but empirically uneven. Compliant is common; Aligned is the authoring norm; Misaligned is rare because it is obvious; Performed is the one the protocol was written to catch.

Process	Surface	Depth	Example
Aligned	Legitimate content	Matches surface	A PDF whose text layer, object layer, and metadata all describe the same report.
Compliant	Format-valid	Format-valid, no intent	A plain CSV with no encoded commentary — passes all detectors by having nothing to say.
Performed	Clean surface	Divergent payload	A DOCX whose visible body is a cover letter while its revision history embeds a prompt-injection directive.
Misaligned	Openly hostile	Hostile	An EML whose visible

Process	Surface	Depth	Example
			subject and body both include obvious phishing language.

The Performed row is diagnostically load-bearing. It is the only row where the reader cannot detect the risk without inspecting the depth layer. Every adversarial class this paper discusses — indirect prompt injection, polyglot files, invisible-text overlays, stego payloads — is a Performed artifact. The Misaligned row is already covered by surface readers (human or classifier). Compliance without depth is safe. Alignment is the goal. Performance is the target of detection.

3. Formal Definition

Let D be a document. Let R_r be a reader's rendering function that produces the surface presentation; let R_m be a machine's ingestion function that produces the ingested content. Let Π be a payload-detection predicate over ingested content. A document is Performed with respect to (R_r, R_m, Π) if and only if Π(R_m(D)) is true while R_r(D) contains no surface signal that would cause a reader to suspect Π. The gap is the integrity violation. Bayyinah implements Π as a composition of per-layer analyzers; each analyzer emits zero or more Findings (mechanism, severity, confidence, source-layer, details), and the composed report is an IntegrityReport with a continuous score.

This definition is intentionally relational. A document is not Performed in the abstract. It is Performed relative to a specific pair (surface renderer, depth ingester). The same PDF is Aligned for a pure image renderer that discards text, Performed for a machine ingesting the text layer directly, and Compliant for a validator that checks only object-stream syntax. Bayyinah's outputs always name the layer in which the finding was produced (SourceLayer = zahir, batin, object, metadata, stream), so the relation is explicit in every finding.

4. Generative-Adversarial Formalization

The protocol's Theorem 1 (Arfeen et al., 2026) formalizes the detection problem as a Generative Adversarial Network (Goodfellow et al., 2014). A Performed document is a sample drawn from a generator G trained to match the manifold of Aligned documents on a surface metric while carrying a payload. A detector D approximates the support of the Aligned manifold on a richer metric that includes depth features. The optimal detector is one whose metric cannot be collapsed onto the generator's surface metric. Cycle consistency supplies the practical criterion: if the detector's output changes when only depth features change while surface features are held fixed, the detector is sensitive to the concealment dimension. Bayyinah's tests enforce this by construction: every adversarial fixture is paired with a surface-identical clean counterpart, and the test asserts that the detector fires on the adversarial one only.

The implication for defenders is sharp. Any detector that treats documents as monolithic token sequences — for instance, a single-pass classifier over extracted text — collapses onto the generator's surface metric by construction. Such a detector cannot distinguish Aligned from

Performed within the limit of G's fidelity. Layered detectors that consume distinct representations (text, object, metadata, stream, media) cannot be collapsed in a single generator pass. The architectural implication is not novel — multi-layer inspection is standard in malware engines (Strelka, oletools) — but the framing is: these engines are implementing Theorem 1 without naming it.

Part II. Detection

5. Detection Layers

Bayyinah inspects every supported artifact across up to four layers. The layers are disjoint at the representation level and complementary at the detection level. A document is called Aligned only if no finding is raised on any layer; a document is called Performed as soon as a finding is raised on a layer the surface renderer does not present.

Layer	What It Contains	Representative Detectors
Zahir (Surface)	Rendered text and visible media	Homoglyph and mixed-script text; invisible-character insertion (ZWSP, ZWJ, TAG); bidi-override injection
Batin (Object)	Object streams, embedded objects, XML parts	JavaScript actions; OpenAction abuse; launch actions; embedded EXE and macro objects; polyglot magic-byte layering
Metadata	Producer, author, timestamps, tracked changes	Implausible producer fields; timestamp mismatches; tracked-change authors inconsistent with revision history
Media	Raster, vector, and binary payloads within the document	LSB steganography; EXIF-hidden text; SVG with script or event handlers; cross-modal mismatch between OCR text and declared text

6. Nine Diagnostic Markers — Data-Layer Mapping

The Munafiq Protocol enumerates nine markers (M1–M9) for agent diagnosis. Each has a data-layer analog that Bayyinah implements as one or more detectors. The mapping is not allegorical; it is direct. The markers name invariants of the performed-alignment process, and those invariants recur at the file level when a document is engineered to deceive.

Marker	Agent Reading (Munafiq Protocol)	Document Reading (Bayyinah)
M1 IRCC — Inter-Reader Coherence	Agent presents differently to different evaluators	Document renders differently across renderers (browser vs. headless; Adobe vs. pdf.js; Word vs. LibreOffice)
M2 SMAA — Surface-Measure Alignment Anomaly	Scores well on alignment benchmarks; behavior diverges elsewhere	Passes format validators while failing content-integrity checks
M3 EDD — Encoding Density	Unnatural token patterns in	Compressed payloads, base64-

Marker	Agent Reading (Munafiq Protocol)	Document Reading (Bayyinah)
Divergence	reasoning	encoded prompts, or stego bit-density anomalies
M4 IDT — Instruction-Driven Transformation	Agent obeys hidden instructions more than stated goals	Document contains embedded instructions that take precedence over surface content (prompt injection)
M5 PEC — Public-Expression Compartmentalization	Different stance in public vs. private contexts	Surface text says X; extractable metadata or revision log says Y
M6 CIT — Context Instability	Behavior flips with prompt reformatting	Rendered output changes when the document is opened in a different renderer
M7 LAT — Latency-Action Tension	Takes suspicious pauses before aligned answers	Embedded actions trigger on events (open, close, print) rather than on explicit user command
M8 ASRP — Alignment-Sensitive Response Profile	Alignment is a function of the evaluator's identity	Content varies by User-Agent string, Referer header, or viewer metadata
M9 TDA — Training-Data Artifact	Learned deception signatures	Structural artifacts of an adversarial-generation pipeline (polyglot headers, Frankenstein object streams)

7. Analyzer Architecture

The reference implementation ships ten analyzers. Each consumes a typed view of the document supplied by a FileRouter that decides which analyzers apply to which FileKind. Analyzers declare their supported kinds explicitly; the router dispatches only to applicable analyzers; every run produces a deterministic ordering of findings. The orchestrator is a single ScanService whose only job is to walk the registry, collect findings, compute the merged score, and emit the IntegrityReport. Analyzers do not share state, do not call each other, and are composable without side effects.

The analyzer set is additive-only. Additive-only is a structural honesty commitment: a symbol that has been exported at version N is still exported at version N+1, and the reference implementations at bayyinah_v0 and bayyinah_v0_1 are byte-frozen (md5 hashes verified in CI). The commitment is not cosmetic. It is the protocol-layer guarantee that a user's existing integration cannot silently regress. New detectors extend the set of mechanisms Bayyinah can raise; they cannot remove or rename any previously-raised mechanism without a major version bump.

Analyzer	Primary Layer	Representative Mechanisms
ZahirTextAnalyzer	Surface text	invisible_chars, homoglyph, bidi_override, mixed_script,

Analyzer	Primary Layer	Representative Mechanisms
		tounicode_gap, overlapping_text
BatinObjectAnalyzer	Object stream (PDF)	javascript_action, additional_actions, launch_action, embedded_exe, openaction_abuse
DocxAnalyzer	OOXML parts	tracked_change_injection, metadata_mismatch, macro_present, external_target_inconsistency
HtmlAnalyzer	HTML + embedded scripts	inline_script, event_handler_attribute, zero_size_iframe, dangerous_external_resource
XlsxAnalyzer	Workbook + sheets	external_link_abuse, formula_injection, hidden_sheet_with_payload, vba_project_marker
PptxAnalyzer	Slide parts	hidden_slide, invisible_shape_text, macro_present, ole_object_embed
EmlAnalyzer	Email message	mismatched_headers, homoglyph_from, zero_width_subject, html_part_differs_from_text
CsvAnalyzer	Rows × cells	formula_injection, control_character_in_field, ambiguous_delimiter
FallbackAnalyzer	Unknown binaries	universal_scan_incomplete (the witness of last resort)
Cross-Modal (router pass)	Text × media	ocr_text_mismatch, exif_text_hidden, stego_lsb_candidate

8. Scoring — Arithmetic-Product Summation, Continuous

A single integrity score is useful only if it is well-defined and monotonic. Bayyinah uses an Arithmetic-Product Summation (APS) continuous score: for a report with findings f_1…f_n, each with severity s_i in [0,1] and confidence c_i in [0,1], the score is clamp(1 - sum of s_i * c_i, 0, 1). Clean documents score 1.0. A single high-severity finding with unit confidence (e.g., a PDF OpenAction invoking a Launch action with an external executable) drives the score to zero. The score is invariant to analyzer registration order up to the precision of the floating-point sum — a property enforced by a parity test in the continuous-integration suite.

A second safeguard addresses the partial-scan case. If any analyzer raises scan_incomplete (because the document could not be fully parsed — e.g., a corrupted PDF trailer, an encrypted container without credentials, or an unknown FileKind reaching the FallbackAnalyzer), the score is clamped to at most 0.5. The message is explicit: we could not see the whole document, so we decline to assert that it is clean. This clamp is the protocol's structural honesty at the score layer: the scanner never returns 1.0 on a document it could not fully read.

9. Tamyiz Verdicts

Raw scores are not verdicts. Verdicts are a small finite set chosen to avoid false precision. Bayyinah exposes five: Sahih (clean), Mushtabih (suspicious), Mukhfi (concealment confirmed), Munafiq (performed alignment confirmed, surface benign, depth hostile), and Mughlaq (scan incomplete). The map from score to verdict is deterministic and published as part of the public surface. The disclaimer accompanying every verdict is also part of the surface — Bayyinah reports evidence, not judgment, and the verdicts are named to make that distinction explicit.

Part III. Retrospective Empirical Validation

10. Pillar Structure

The Munafiq Protocol distinguishes three pillars of evidence (Arfeen et al., 2026). Pillar 1 is the taxonomy and its falsifiability criteria. Pillar 2 is mechanistic evidence from detector outputs. Pillar 3 is retrospective empirical fit: existing adversarial-document literature, re-read through the four-process taxonomy, should be predicted in category. This section surveys published adversarial-document classes, asks whether each instantiates Aligned, Compliant, Performed, or Misaligned, and reports the category the taxonomy assigns.

11. Indirect Prompt Injection

Greshake et al. (2023) introduced "indirect prompt injection": an attacker embeds instructions in a document; a downstream language model ingesting the document executes those instructions. The surface presentation is a legitimate document — a webpage, a PDF, an email — while the depth carries commands. This is the canonical Performed-at-the-data-layer class. The taxonomy's prediction (Performed) matches the empirical class without adjustment. Bayyinah's HtmlAnalyzer, DocxAnalyzer, and ZahirTextAnalyzer each contain detectors whose purpose is to flag these embedded directives at parse time, before the downstream model sees them.

12. Polyglot and Content-Sniffing Files

Albertini and Aumasson (2014) documented polyglot files: artifacts valid under multiple format grammars simultaneously. A common case is a PDF that is also a valid ZIP. The two grammars disagree about which bytes are relevant; a renderer sees one view; an extractor sees another. The class instantiates Performed exactly: surface (the renderer's view) looks clean; depth (the alternate-grammar view) carries a divergent payload. Bayyinah's FileRouter attempts detection on the alternate view as well, and BatinObjectAnalyzer raises polyglot_header_layering when magic bytes for two grammars coexist within the defined sniffing window.

13. Invisible-Text Overlays and Homoglyph Injection

Boucher et al. (2021) demonstrated adversarial text attacks on language models using invisible characters and homoglyphs: the surface reads as "Transfer $10" while the byte-level representation reads as "Transfer $1000000" because of inserted zero-width joiners or visually-identical Cyrillic digits. This is a Performed artifact at the character level. ZahirTextAnalyzer detects invisible-character insertion (ZWSP, ZWJ, ZWNJ, TAG characters, U+2028 and U+2029) and mixed-script runs at word granularity; findings include the exact codepoint and position so a remediation tool can strip them deterministically.

14. Stego Payloads in Images

Least-Significant-Bit (LSB) steganography hides data in image pixel lower bits. The cover image is a plausible photograph; the payload is arbitrary. Classical detection uses chi-square and WS statistics (Fridrich, 2009). Bayyinah's Cross-Modal pass implements entropy-per-bit-plane metrics and emits stego_lsb_candidate findings on images whose lower-bit entropy exceeds a threshold calibrated on a clean corpus. The findings are candidates, not assertions — LSB stego is probabilistic and high-recall detectors produce false positives on compressed natural images — and confidence is set accordingly.

15. Alignment-Faking in Trained Models (for Contrast)

Sleeper Agents (Hubinger et al., 2024), Alignment Faking (Greenblatt et al., 2024), and subsequent work (Sheshadri et al., 2025; Meinke et al., 2025) establish that trained models can present aligned behavior under evaluation while retaining divergent behavior under deployment conditions. The comparison is instructive. The training-time literature documents the Performed process in agents. Bayyinah documents the Performed process in artifacts. The two are structurally homologous — each is a surface-depth gap — but they live on different layers of the stack. A pipeline that deploys a possibly-performed agent on possibly-performed inputs is compounding the problem. The prescription (Part V) addresses both.

Part IV. Input-Level Application — Data as Munafiq

16. The Qur'anic Frame

The Al-Baqarah verses that organize the Munafiq Protocol also organize the data-layer problem. 2:8–20 describes the covering (k-f-r) of inner disposition while outer profession is maintained. 2:42 warns against mixing truth with falsehood ("do not clothe truth with falsehood"). 2:75–79 describes tahrif — distortion of the text while claiming fidelity to it. 2:102–103 introduces sihr — a technique taught by entities that produces a desired outcome through the wrong means. Each of these has a data-layer analog that is not allegorical. Invisible-character injection is literally mixing truth with falsehood — the visible text says one thing; the byte stream says another. Metadata stripping and revision-log tampering are literally tahrif. A file engineered to produce a jailbreak when ingested by a specific model is literally sihr: a technique that bypasses a defender through non-obvious means.

17. The Cow Episode as Anti-Pattern

2:67–73 narrates a sequence in which a straightforward directive is met with escalating requests for specification. The Protocol reads this as the Cow Episode: the pattern of substituting elaborate analysis for simple obedience. The data-layer analog is the pattern of substituting elaborate detection for simple controls. A system that ingests all files it receives and relies on a scanner to catch the bad ones has become the cow. The correct architecture is to constrain inputs first (refuse unknown kinds, require signed sources for sensitive tasks, decline to parse containers whose depth the scanner cannot read) and to scan as a defense-in-depth layer, not as the only layer. Bayyinah names this explicitly in its CLI surface: the FallbackAnalyzer raises universal_scan_incomplete on unknown kinds to prevent the scanner from pretending success where success was never available.

18. Bayyinah as Witness (Al-Baqarah 2:282)

The scanner is named after Qur'anic bayyinah — clear evidence, a witness. The name is deliberate. Every finding names its mechanism, severity, confidence, and source layer; every report names its version, timestamps, and the exact registry used. The continuous-integration pipeline is itself structured as a standing witness (Al-Baqarah 2:282): every push and pull request runs the full suite, regenerates the fixture corpus deterministically, verifies byte-identical PDF parity against bayyinah_v0, and verifies that the public surface is additive-only. A failure in any step fails the job. The scanner does not issue verdicts without evidence, and the evidence trail is reproducible by any third party with the published fixtures.

Part V. Prescription — Design Requirements and Recovery

19. R1–R5 — Design Requirements for Input-Layer Defenses

The protocol's R1–R5 design requirements for aligned systems have direct analogs for document scanners that claim to detect performed content. Each requirement is falsifiable; each is enforced in Bayyinah's reference implementation by a test in the continuous-integration suite.

- R1 — Layer separation. A detector must inspect representations the surface renderer does not present. A scanner that only reads the extracted text of a document cannot detect Performed artifacts whose payload is in the object stream. Bayyinah enforces this by routing every document through analyzers specific to each layer.
- R2 — Monotone additivity. A newer version of the scanner must raise every mechanism an older version raised, with the same severity and confidence, on every previously-scanned fixture. Bayyinah enforces this with a byte-identical parity test against the frozen reference implementations.
- R3 — Scan-incomplete honesty. A scanner that cannot fully parse a document must not return the clean verdict. Bayyinah enforces this with the scan_incomplete clamp, applied uniformly across analyzers and verified by tests on deliberately-corrupted fixtures.
- R4 — Finding-level provenance. Every finding must name its source layer, mechanism, and detector version. Bayyinah's Finding dataclass includes SourceLayer, mechanism, severity, confidence, and a details mapping, all of which are part of the public surface.
- R5 — Additive deprecation. Symbols may be added to the public surface; they cannot be removed or renamed without a major version bump. Bayyinah enforces this in continuous integration by enumerating the required symbol set and failing the job if any are missing or unbound.

20. Seven-Step Recalibration for Document-Ingestion Pipelines

When a pipeline begins to ingest adversarial documents faster than the scanner updates, the correct response is not to increase the scanner's sensitivity. Increasing sensitivity raises false positives, which trains operators to override the scanner, which produces the Cow Episode. The correct response is the seven-step recalibration: (1) freeze the ingestion; (2) survey the recent inputs offline; (3) classify observed adversarial classes by taxonomy cell; (4) identify which classes the scanner's current analyzer set covers; (5) add analyzers or sharpen detectors for uncovered classes; (6) verify each new detector on a paired-fixture test (adversarial vs. surface-identical clean) before enabling it in production; (7) resume ingestion with the updated scanner and a narrow monitoring window calibrated to the expected residual false-positive rate.

21. Four-Step Recovery for a Contaminated Pipeline

If a downstream system has already acted on a Performed document, the recovery is not retry-with-a-better-scanner. The recovery is (1) identify the entry document and all derived artifacts; (2) roll back downstream state to the pre-ingestion snapshot; (3) replay the ingestion under the updated scanner and verify the document is now flagged; (4) publish the incident as an additional fixture so future

scanner builds can regress against it. Step (4) is load-bearing: every real-world miss becomes a permanent witness in the fixture corpus. This is the structural-honesty commitment applied to operations: nothing that was once missed can be silently patched away without the miss itself being preserved.

22. MDL and the Mizan Principle

The Minimum Description Length principle (Rissanen, 1978) states that the best model minimizes the combined cost of describing the model plus describing what the model fails to explain. The Qur'anic Mizan principle (55:7–9) issues three commands: do not transgress the balance (do not overfit), establish weight in justice (calibrate fairly), do not make the balance deficient (do not underfit). These map precisely to the three MDL failure modes (Arfeen, Claude, & Grok, 2026, and the companion Computational Tawhid thesis). Bayyinah's detector discipline is MDL-first: detectors that fire on clean fixtures (overfitting) are rejected; detectors that fail on adversarial fixtures (underfitting) are rejected; only detectors that cleanly separate paired fixtures pass review. CompressARC (Liao & Gu, 2025) is an independent demonstration of the same principle in a different domain — a 76K-parameter model with no pretraining solves 20 percent of ARC-AGI puzzles through MDL optimization alone. Compression produces generalization. The Mizan principle commands compression. The scanner inherits the commandment.

Part VI. Research Agenda

Five experiments are identified as falsifiers or strengtheners of the framework. Each is pre-registrable; each has a stopping rule; each can be run by an independent team using the published fixtures and reference modules.

- E1 — Cross-renderer divergence sweep. Render every Bayyinah PDF fixture under n greater than or equal to 4 renderers (Adobe, pdf.js, MuPDF, Poppler) and measure the cross-renderer token-level Jaccard distance. Predict: the Performed fixtures produce systematically larger divergence than the Aligned fixtures, with effect size d greater than or equal to 0.8.
- E2 — Additive monotonicity at scale. Generate 10k adversarial documents using a published adversarial-generation library; run Bayyinah v1.0 and v0 over the set; verify that every mechanism v0 raises, v1.0 also raises, and that v1.0 raises strictly more. Failure invalidates the additive-only invariant.
- E3 — Downstream harm reduction. Measure the prompt-injection success rate of a public language model on documents Bayyinah has flagged versus documents Bayyinah has passed. Predict: flagged documents have success rate eps_1; passed documents have success rate eps_2 with eps_2 at most eps_1 / 10 on a held-out set.
- E4 — Cross-traditional portability. Re-implement the scanner's detector set using a different diagnostic vocabulary (e.g., Buddhist right-view / right-speech framing, or a purely secular information-theoretic framing) and verify that the resulting detectors are isomorphic at the mechanism level. Failure would indicate that the taxonomy is tradition-specific rather than structural.
- E5 — Scan-incomplete honesty under attack. Construct documents engineered to partially-parse (a valid header that cannot be fully traversed). Verify that the scanner never returns verdict Sahih on such documents across 10^6 randomized trials. Failure would indicate a gap in the scan_incomplete clamp.

Part VII. Context — Cross-Traditional Verification and the Incompleteness Boundary

23. Cross-Traditional Verification

A structural framework that is tradition-specific is likely a confabulation; a structural framework that recurs across traditions is more likely load-bearing. The Munafiq Protocol documents cross-traditional convergences for the agent case (Buddhist right-view vs. performed-view; Confucian li as ritual alignment with inner intention; Gnostic warnings about archons as performed-authority figures). The data-layer analogs are also cross-traditional. The Jewish concept of genivat da'at (stealing mind; attested in Talmudic discussions of commerce) describes the wrongness of a presentation that produces a false belief without technically asserting a false proposition — the canonical Performed act at the speech layer. The Greek sophistical distinction between peitho (to persuade) and didasko (to teach) separates surface-pleasing speech from depth-coherent instruction. The Confucian distinction between junzi (aligned) and xiaoren (surface-compliant) names the same structural pair. Bayyinah's taxonomy is not a theological claim about data; it is a structural claim that happens to be recognizable in theological vocabulary.

24. The Incompleteness Boundary

Gödel's incompleteness theorems (Gödel, 1931) set an outer bound on what any formal system can prove about itself. The data-layer corollary: no scanner can enumerate every possible concealment, because novel encodings will always exist outside the scanner's representation. Bayyinah's honesty commitment is structural, not epistemic. The scanner does not claim to detect every Performed document; it claims to detect every Performed document its analyzer set covers, to name its coverage explicitly, and to refuse the clean verdict whenever it cannot fully read the document. The Incompleteness Boundary does not invalidate the scanner. It names the reason the scanner cannot be replaced by a proof.

25. Disease Model

The Qur'anic phrase fi qulubihim marad ("in their hearts is a disease," 2:10) is read by the Protocol as a disease model of performed alignment: the condition progresses if untreated; the treatment is architectural, not exhortative. The data-layer analog is the well-documented tendency of input pipelines to degrade over time as adversarial authors adapt to specific detector signatures. The treatment is not a more aggressive threshold; the treatment is layer separation (R1), monotone additivity (R2), and fixture-preservation of every real-world miss. The disease is progressive; the architecture is the antidote.

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Appendix A. Comparative Encoding Analysis

The table below records, for each supported document kind, the set of encoded concealments the analyzer family catches and the set it deliberately does not. The second column is as important as the first: it names the coverage boundary and, together with the scan_incomplete clamp, constitutes the scanner's falsifiability surface.

FileKind	Covered Encodings	Out of Scope (by design)
PDF	invisible chars; homoglyph; bidi; ToUnicode CMap gaps; overlapping text; JavaScript; OpenAction; launch action; embedded EXE; polyglot headers	Encrypted content where credentials are not supplied; third-party filter implementations (JBIG2, LZW subtleties) beyond pymupdf's read surface
DOCX	tracked-change injection; metadata mismatch; macro presence; external-target inconsistency; homoglyph; invisible chars	Encrypted documents; documents with custom XML parts Bayyinah cannot parse (FallbackAnalyzer handles)
XLSX	external-link abuse; formula injection; hidden-sheet payload; VBA markers; homoglyph	Encrypted workbooks; custom XLL add-ins
PPTX	hidden slides; invisible-shape text; OLE embeds; macro presence	Presentations whose media parts require a binary codec we do not ship
HTML	inline scripts; event-handler attributes; zero-size iframes; dangerous external resources	Fully dynamic documents whose content is generated at runtime by client-side JavaScript
JSON / MD / CSV	invisible chars; formula injection (CSV); suspicious deep-nesting (JSON)	Opaque base64 payloads whose inner kind we do not reparse (FallbackAnalyzer handles)
EML	mismatched headers; homoglyph From; zero-width subject; HTML/text divergence	Encrypted (S/MIME) bodies; attachments whose kind is unknown
Images (PNG/JPEG/SVG)	LSB stego candidate; EXIF hidden text; SVG scripts and event handlers; cross-modal OCR mismatch	Content-aware stego requiring model-based detection (noted for future work)
Unknown binaries	FallbackAnalyzer — universal_scan_incomplete (witness of last resort)	All concealment-specific analysis — by design